Research Note

S&P 500: The AI Wave Market Value, Fundamentals and Valuation

May 2023 - October 2026

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Key Thesis

AI-related companies' share of market capitalization rose from 31.2% to 47.4%, while their share of company revenue rose from 14.4% to 20.0%.

Market value and fundamentals expanded at different rates.

GAAP earnings include investment revaluation gains, and classifications are retrospective thematic labels. This study does not identify AI's causal contribution.

Main classification: 41 broad AI reference companies plus seven historical exits. Only contemporaneous index members are included at each observation.

Research Summary: Fundamentals, Pricing and Reinvestment

The AI group's market capitalization rose 191.0%, versus 68.6% growth in company revenue. Revenue growth clearly outpaced the rest of the index, but market value expanded considerably faster. Changes in constituents and share counts also affect aggregates; the entire gap cannot be attributed to valuation.

GAAP net income rose 233.4%, and aggregate P/E fell from 33.8 to 29.5 times, while P/S rose from 5.0 to 8.6 times. Higher profit margins and non-operating gains create divergent valuation signals.

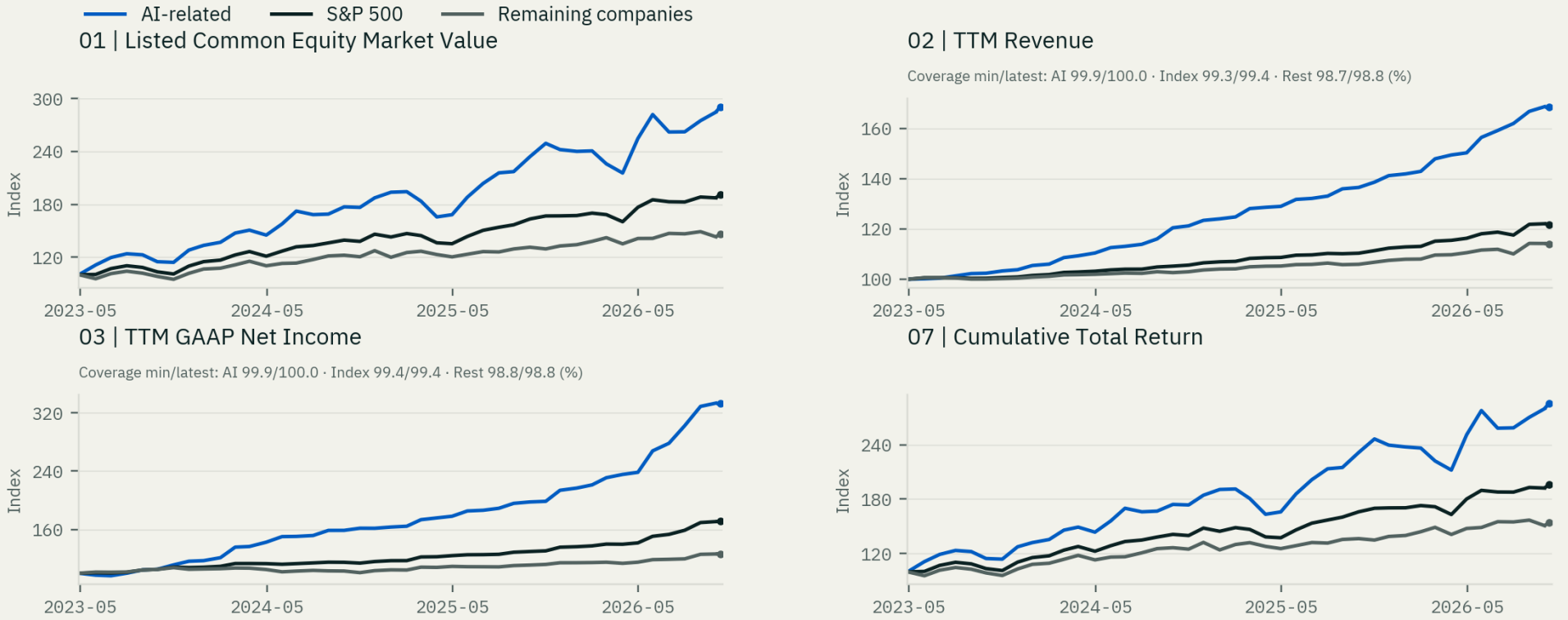
The latest 41 AI companies represent 8.2% of companies and 47.4% of market capitalization. The five largest account for 63.9% of AI market value. Over the past year, 61.0% of AI companies rose, versus 62.9% of the remaining companies. Aggregate outperformance does not imply broad gains within the group.

Capital expenditure / revenue rose from 10.9% to 18.8%, while the free cash flow margin fell from its within-period peak of 18.3% to 14.8%. This reveals tension between reinvestment intensity and cash realization; it does not yet establish whether investment returns are good or poor.

Base to latest	AI-related	S&P 500	Remaining
Market value change	+191.0%	+91.4%	+46.2%
TTM revenue change	+68.6%	+21.8%	+13.9%
TTM GAAP income change	+233.4%	+71.1%	+26.7%
Cumulative total return	+185.7%	+96.3%	+54.2%
Aggregate P/E	33.8 → 29.5	23.3 → 25.9	20.4 → 23.3
Non-financial P/S	5.0 → 8.6	2.3 → 3.7	1.7 → 2.2
Capex / revenue	10.9% → 18.8%	5.8% → 8.5%	4.7% → 5.1%
FCF / revenue	12.5% → 14.8%	8.8% → 10.2%	8.0% → 8.7%

Ratios use matched samples. Company revenue is not AI business revenue. Return weighting differs between the groups and the index.

Market Value and Fundamentals: Sources of Growth

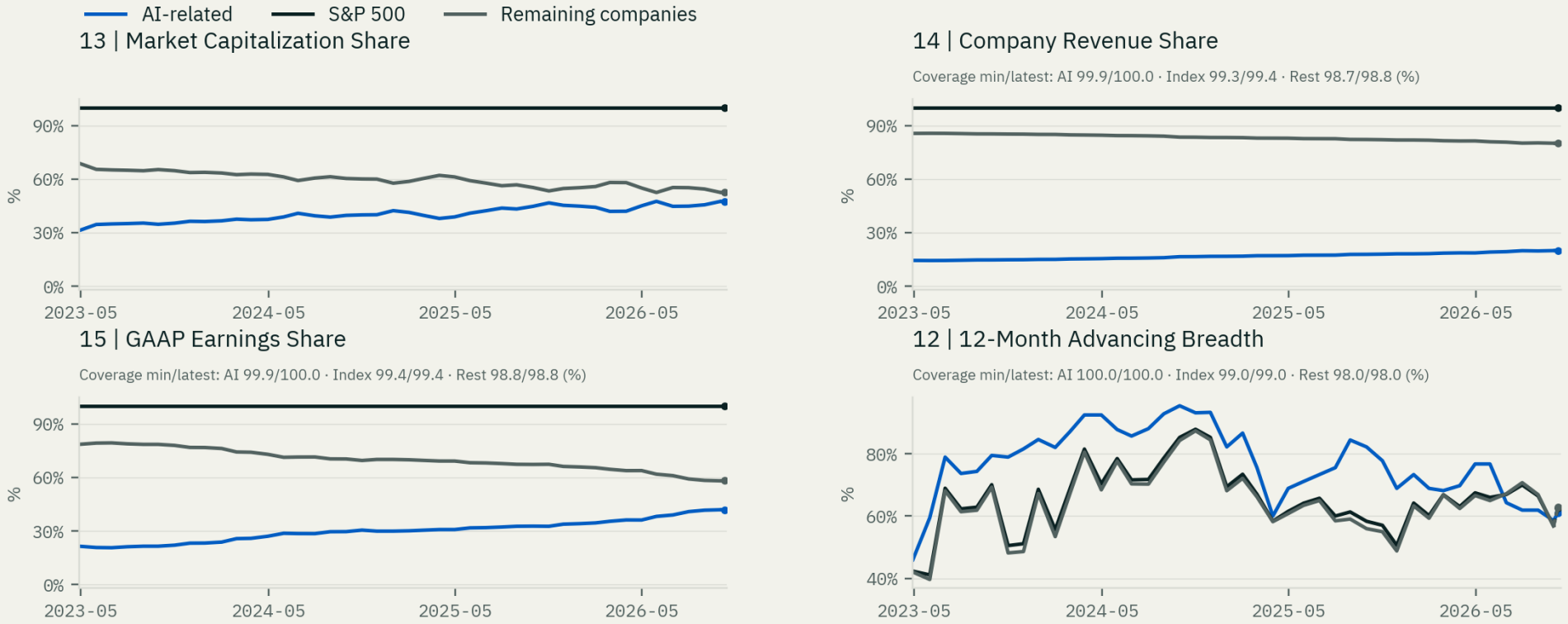


Facts: AI revenue rose 68.6%, versus 13.9% for the remaining companies. AI market value rose 191.0% and GAAP earnings 233.4%. Both fundamentals and market value diverged substantially.

Interpretation: Reported earnings provide accounting support for the gains, but GAAP earnings include investment revaluations. Aggregates for changing constituents also reflect composition effects and cannot directly identify AI's causal contribution.

Sources: User-provided CRSP / Compustat, Massive and SEC. Base: 2023-04-28 = 100. Data through 2026-10-09; dashed segments denote the October snapshot.

Shares and Breadth: Concentration Is Not Broad Participation

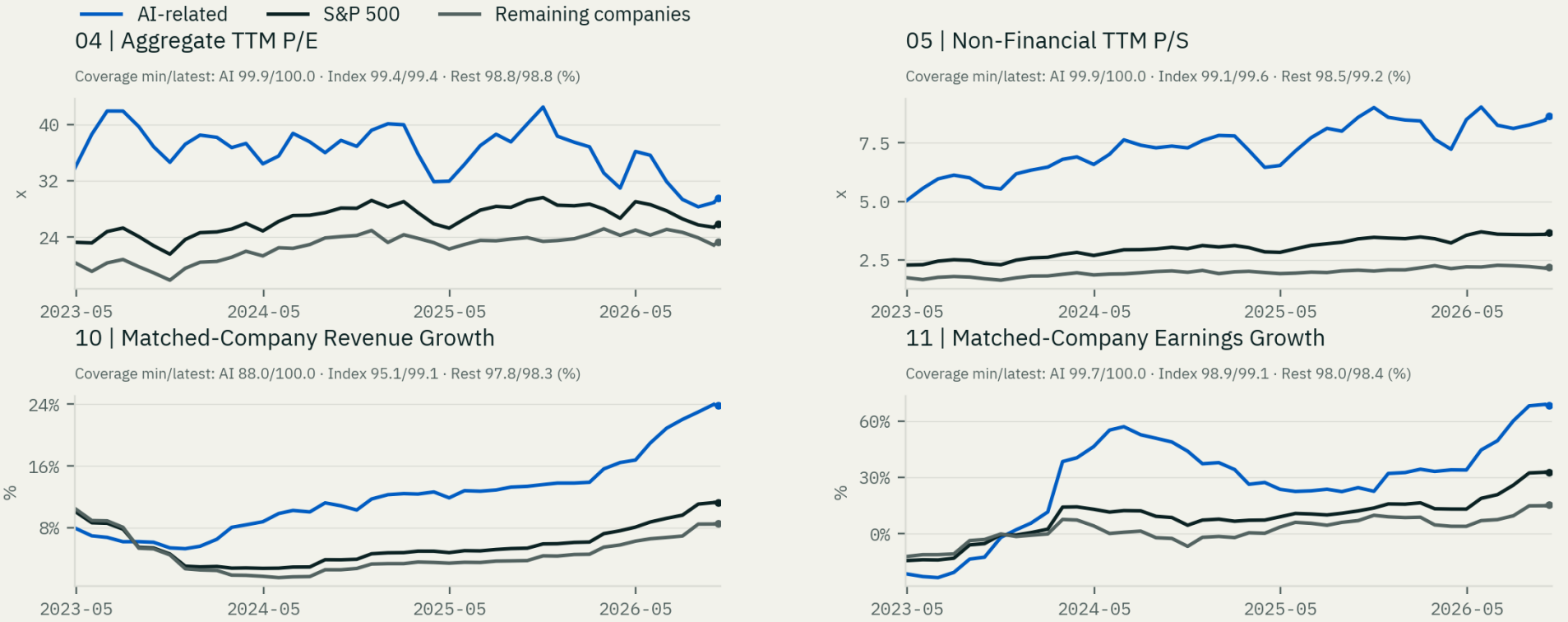


Facts: AI market value share rose from 31.2% to 47.4%, revenue share from 14.4% to 20.0%, and earnings share from 21.5% to 41.8%. Latest advancing breadth is 61.0%, below the remaining group's 62.9%.

Interpretation: Market value is increasingly concentrated in thematic companies. The five largest account for 63.9% of AI market capitalization. These charts cannot yet determine how many companies drive the growth; company-level contribution analysis is needed.

Sources: User-provided CRSP / Compustat, Massive and SEC. Base: 2023-04-28 = 100. Data through 2026-10-09; dashed segments denote the October snapshot.

Valuation and Growth: P/E and P/S Send Different Signals

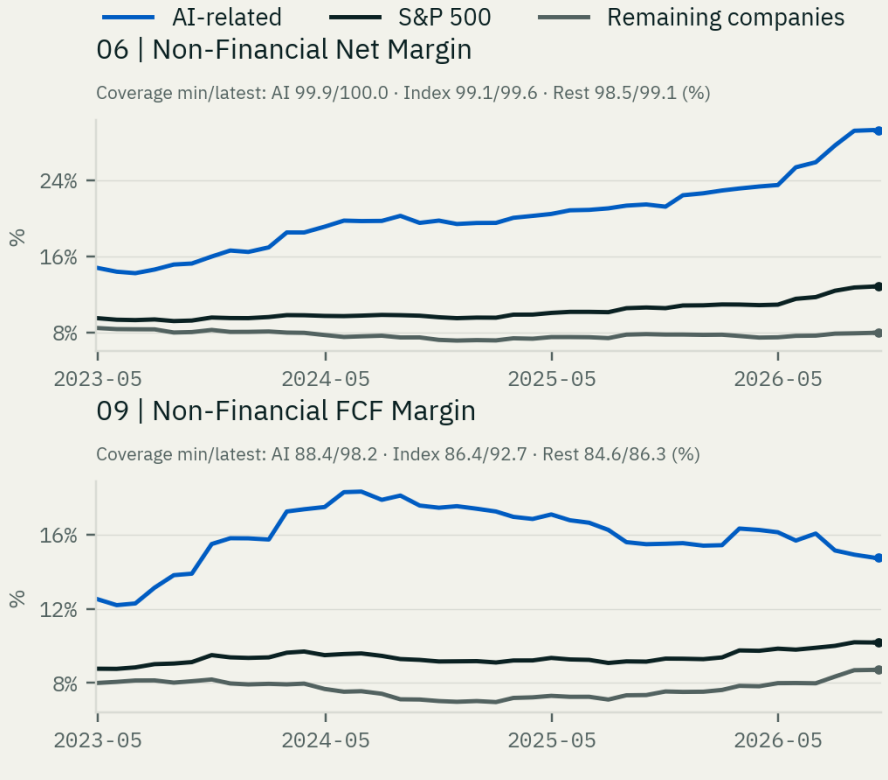


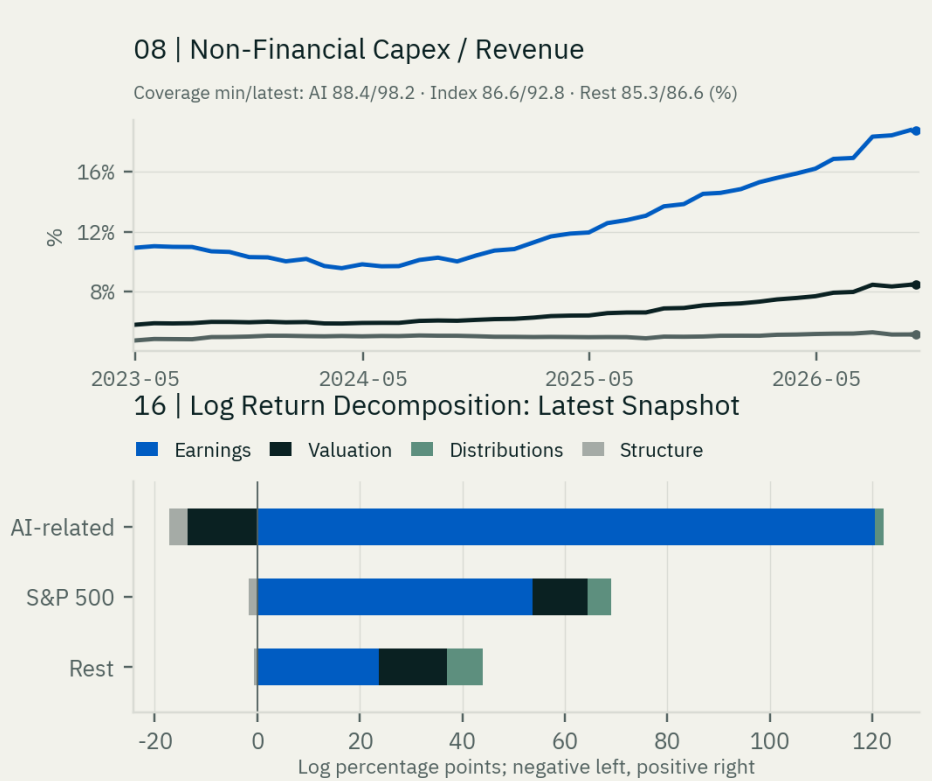
Facts: AI P/E fell from 33.8 to 29.5 times, while P/S rose from 5.0 to 8.6 times. Latest matched-company revenue growth is 23.8% year on year, versus 8.6% for the remaining group.

Interpretation: Pricing relative to revenue increased, while pricing relative to GAAP earnings fell. This is not evidence that valuations rose across the board or that the group is now cheap. Alphabet's Q2 investment gains increased net income by $77.1 billion.

Sources: User-provided CRSP / Compustat, Massive and SEC. Base: 2023-04-28 = 100. Data through 2026-10-09; dashed segments denote the October snapshot.

Earnings and Cash: The Reinvestment Behind Expansion





Facts: AI net margin rose from 14.8% to 29.3%, and capital expenditure / revenue from 10.9% to 18.8%. The FCF margin fell from 18.3% in June 2024 to 14.8%, still above its initial 12.5%.

Interpretation: Accounting earnings and cash generation can diverge. Higher reinvestment intensity coincides with a lower FCF margin, but the entire decline cannot be attributed to AI capital expenditure before fixed-sample and company-level tests are completed.

Conclusion

The central finding is that, since April 2023, the market capitalization of 41 AI-related S&P 500 companies rose 191%, alongside 68.6% revenue growth and 233.4% GAAP net income growth. Their market value share increased from 31.2% to 47.4%, and their earnings share from 21.5% to 41.8%. The logarithmic decomposition attributes about 120 percentage points of cumulative return to earnings growth, with a negative valuation contribution of roughly 13 percentage points. This suggests that realized earnings, rather than multiple expansion alone, were the main accounting driver of the rise. Aggregate P/E fell from 33.8 to 29.5 times, which does not align neatly with a bubble narrative. However, almost half of the earnings expansion came from a doubling of net margin, from 14.8% to 29.3%, rather than revenue growth itself. The case for reasonable valuation therefore depends heavily on the sustainability of those margins.

Earnings quality and concentration are two reasons to apply a discount to that interpretation. First, GAAP earnings contain substantial non-operating investment gains. Mechanically excluding Alphabet's $77.1 billion equity investment gain in a single quarter raises the AI group's P/E from 29.5 to 31.7 times. Meanwhile, P/S rose from 5.0 to 8.6 times, indicating materially higher pricing of revenue; a lower P/E does not necessarily mean the group became cheaper. Second, while net margin doubled, the FCF margin rose only from 12.5% to 14.8% and fell from its mid-2024 peak of 18.3%. Together with the rise in capital expenditure / revenue from 10.9% to 18.8%, this indicates a marked decline in the conversion of accounting profits into cash, characteristic of a heavy investment phase. The five largest companies account for 63.9% of group market value, and advancing breadth of 61.0% is below the remaining companies' 62.9%. The AI group's performance may therefore be driven mainly by a few giants. Taken together, we see real fundamental support for the AI theme, but market prices already embed high expectations for sustained margins and the eventual returns on capital expenditure. Until tests excluding the largest companies, operating-earnings adjustments and cash-return analysis are completed, the more measured stance is to recognize the theme while remaining cautious about claims that valuation is reasonable.

Sources: User-provided CRSP / Compustat, Massive and SEC. Base: 2023-04-28 = 100. Data through 2026-10-09; dashed segments denote the October snapshot.

Methodology, Classification Comparisons and Boundaries

Broad AI plus historical exits: 41 companies; market capitalization share 47.4%; revenue share 20.0%; cumulative research portfolio return 185.7%.

Fixed set of 41 broad AI companies: 41 companies; market capitalization share 47.4%; revenue share 20.0%; cumulative research portfolio return 186.8%.

Fixed set of 29 direct AI companies: 29 companies; market capitalization share 46.0%; revenue share 18.5%; cumulative research portfolio return 190.8%.

The study runs from May 2023 through October 9, 2026, with April 28, 2023 as the base date and complete month-ends through September 2026. Historical membership changes with effective index events, and share classes are consolidated at company level. Changes in share counts and constituents affect market value and fundamental aggregates.

The main classification consists of 41 broad reference companies, supplemented by seven historical exits: ANSS, JNPR, EPAM, TTD, DXC, DAY/CDAY and PAYC. Supplementary labels apply only after evidence of commercial activity becomes effective and while the company is actually in the index. The reference list is a retrospective label; the remaining companies are not necessarily without AI exposure.

Market capitalization covers listed common equity, rather than official free-float value. P/E includes losses and GAAP non-operating gains or losses. P/S, net margin, capital expenditure / revenue and FCF margin use matched non-financial samples. Figure 16 presents the latest cumulative logarithmic decomposition in this compact edition; the standalone figure retains the full monthly four-panel series.

Latest market-cap-weighted coverage for AI revenue / net income is 100% / 100%, versus approximately 99.4% / 99.4% for the full index. AI capital expenditure / revenue and FCF margin coverage is approximately 98.2% / 98.2%, versus 92.8% / 92.7% overall. Each chart states minimum / latest coverage. Classification, exits, missing fields and differences between sources are detailed in the accompanying workbook / audit files; the original pages 25 and 26 are no longer shown separately.

AI and remaining-company portfolios use listed-common-equity market-cap weights with monthly rebalancing; the full-index benchmark is the official S&P total return index. Weights differ and transaction costs are excluded. Matched-company year-on-year growth pairs current members' earlier and later TTM values. Historical earnings expectations are excluded as requested.

Earnings Quality, Sources and Further Validation

Alphabet's original Q2 2026 release reports that net gains on equity securities increased net income by $77.1 billion. Mechanically excluding only this item raises the AI group's latest P/E from about 29.5 to 31.7 times. This is a single-item sensitivity illustration, not a complete adjustment to operating earnings. The GAAP net income series cannot be treated directly as a series of core AI operating earnings.

The existing charts support simultaneous growth in thematic companies' revenue, earnings and market value, alongside higher reinvestment intensity. They do not yet establish how much growth came from AI business, whether the findings survive excluding the largest companies, or whether eventual cash returns on investment will meet market expectations.

Three tests take priority: decompose company-level contributions to revenue, earnings and returns to assess concentration; add operating-earnings and non-operating-item adjustments; and separate direct AI, power and equipment, matching capital expenditure, operating cash flow and returns. These are directions for further research, not analyses completed in this edition.

Original classification reference

Historical membership seed and official checks

SEC Company Facts

Alphabet Q2 original earnings-quality release

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